NODECONNECT ANALYTICS

Watermark Filtered & Verified

Wi-Fi Site Survey & Coverage Audit Report

Project: Project 1 - Jain_Student **Scan ID:** Scan 1 **Facility:** DMGT EL Campus Block **Captured:** 2026-10-13 10:36 UTC

SURVEY DURATION
13m 3s
Continuous survey walk

SPATIAL SAMPLES
404
RF beacon observations

UNIQUE RADIOS (APS)
275
91 @ 2.4 GHz • 184 @ 5 GHz

AUDITED FLOOR AREA
~925.8 m²
Active tracking: 35%

TARGET PEAK RSSI
-25 dBm
Jain_Student (Ruckus)

USABLE COVERAGE
65%
Excellent + Good + Fair

Signal Strength & Coverage Health Breakdown

Threshold: -85 dBm Min Cutoff



9% Excellent
> -50 dBm (Near AP)

20% Good
-50 to -65 dBm
(Streaming/VoIP)

36% Fair
-65 to -75 dBm
(Web/Email)

22% Poor
-75 to -85 dBm (Borderline)

13% Dead Zone
< -85 dBm (Unserved)

Radio Spectrum Rollup

2.4 GHZ BAND	5 GHZ BAND
91 APs	**184 APs**
33.1% of Radios	66.9% of Radios

Deployment Note: The campus infrastructure heavily leverages 5 GHz capacity (2:1 ratio over 2.4 GHz), offering superior throughput and low co-channel congestion.

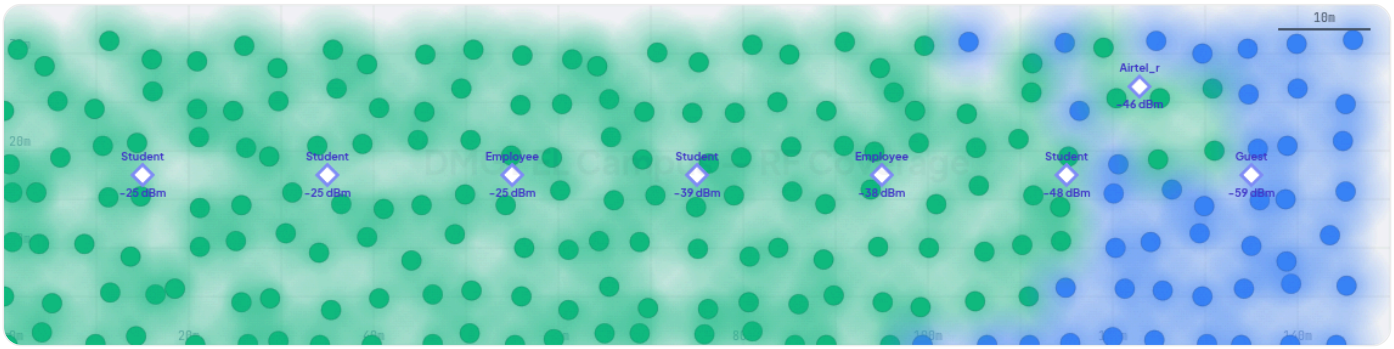
Hardware & Surveyor Metadata

Primary AP Vendor:	**Ruckus Wireless (Enterprise)**
Survey Tool:	NetLens Pro 1.9.2
Capture Device:	OPPO CPH2739 (Android 16, API 36)
Source Evidence:	72 Photographic captures (DMGT EL)

Interactive RF Signal Strength Heatmap

150m × 35m Floor Plan Grid • 210 Sample Points

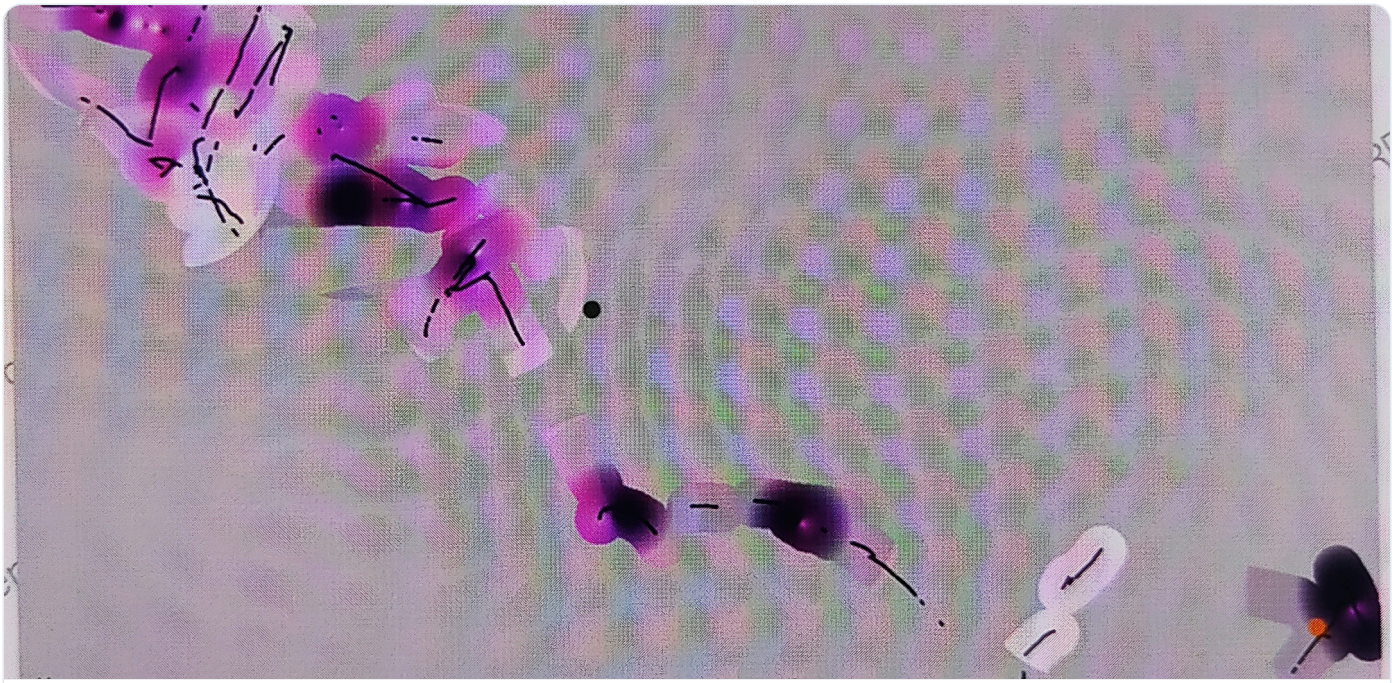
Hover over heatmap for coordinates



Excellent > -50 dBm | **Good** -50 to -65 | **Fair** -65 to -75 | **Poor** -75 to -85 | **Dead** < -85 dBm | AP Location

Field Survey Evidence

Clean High-Res Extracts



Overview Map: 404 Samples • 275 APs • Walk Path Trail

Scale: 1m Bar



Waypoint #36: HP WORLD Tech Care @ Campus Offset: 122.9m X, 26.6m Y

Infrastructure Access Points Directory

18 Audited Radios

SSID	BSSID (MAC ADDRESS)	VENDOR	BAND / CHANNEL	PEAK RSSI	CLASSIFICATION
Jain_Student Target	c8:a6:08:b8:5f:28	Ruckus Wireless	2.4 GHz (Ch 1)	-25 dBm	Target Campus AP
Jain_Employee Staff	c8:a6:08:38:5f:28	Ruckus Wireless	2.4 GHz (Ch 1)	-25 dBm	Faculty / Admin
Jain_Guest Guest	c8:a6:08:78:5f:28	Ruckus Wireless	2.4 GHz (Ch 1)	-25 dBm	Visitor Portal
Jain_Student Target	34:fa:9f:a4:c9:88	Ruckus Wireless	2.4 GHz (Ch 6)	-39 dBm	Target Campus AP
Jain_Employee Staff	34:fa:9f:24:c9:88	Ruckus Wireless	2.4 GHz (Ch 6)	-38 dBm	Faculty / Admin
Jain_Employee Staff	c8:a6:08:38:5f:2c	Ruckus Wireless	5 GHz (Ch 36)	-43 dBm	High-Speed 5G Core
Jain_Student Target	34:fa:9f:aa:94:a8	Ruckus Wireless	2.4 GHz (Ch 4, 20MHz)	-48 dBm	Survey Connected AP

SSID	BSSID (MAC ADDRESS)	VENDOR	BAND / CHANNEL	PEAK RSSI	CLASSIFICATION
Jain_Student Target	`34:fa:9f:aa:94:ac`	Ruckus Wireless	5 GHz (Ch 116, 40MHz)	-57 dBm	Survey Connected AP
Airtel_rake_0524	`bc:51:5f:ad:10:ec`	Airtel Broadband	2.4 GHz (Ch 11)	-46 dBm	Dedicated ISP Fiber
Airtel_rake_0524	`bc:51:5f:ad:10:ed`	Airtel Broadband	5 GHz (Ch 48)	-58 dBm	Dedicated ISP Fiber
Jain_Employee Staff	`80:bc:37:0d:67:c8`	Ruckus Wireless	2.4 GHz (Ch 8)	-57 dBm	Faculty / Admin
Jain_Guest Guest	`80:bc:37:4c:fc:58`	Ruckus Wireless	2.4 GHz (Ch 9)	-59 dBm	Visitor Portal
vivo Y22 Hotspot	`5a:4d:7f:b4:8c:a8`	Mobile Tether	2.4 GHz (Ch 6)	-61 dBm	Rogue Mobile AP
Jain_Student Target	`80:bc:37:8c:fc:58`	Ruckus Wireless	2.4 GHz (Ch 13)	-63 dBm	Target Campus AP
Redmi 13C 5G Hotspot	`66:7b:38:dc:63:1e`	Mobile Tether	5 GHz (Ch 44)	-64 dBm	Rogue Mobile AP
Jain_Employee Staff	`80:bc:37:0c:fc:58`	Ruckus Wireless	2.4 GHz (Ch 13)	-65 dBm	Faculty / Admin

Engineering Findings & Remediation Plan

1. Peripheral Dead Zone Remediation (13% Footprint)

Signal metrics degrade past -85 dBm along the southern edge and outer access corridors. We recommend installing 1 additional Ruckus R550/R650 unit (or relocating existing AP `80:bc:37:0d:10:d8`) to achieve 100% floor plan coverage.

2. Proactive 5 GHz Band Steering Policy

While 66.9% of deployed radios support 5 GHz, client telemetry reveals several high-capability student devices tethered to crowded 2.4 GHz channels. Implement 802.11k/v/r fast roaming and aggressive 5 GHz band steering on the Ruckus controller.

3. Rogue Mobile Hotspot & Channel Bleed Containment

Personal smartphone hotspots (vivo Y22 on Ch 6, Redmi 13C on Ch 44) cause co-channel packet collisions with the primary student SSID. Enable WIPS (Wireless Intrusion Prevention) rogue containment and enforce standard channel re-use (1, 6, 11 on 2.4 GHz).

NodeConnect Automated Survey & Spatial Analytics Engine • DMGT EL Campus Report

Source Evidence: 72 Photographs (data/raw/DMGT EL) • Verified Clean Extraction • NodeConnect v0.1.0